Using direct media access (DMA) controllers can boost responsiveness as well. This hardware device allows input-output devices to directly access the memory with less participation of the processor. It's like dividing the work among people to boost the overall speed, and that is exactly what happens here.

Intel Movidius Myriad X, for example, is specifically a vision processing unit that has a dedicated neural compute engine that directly interfaces with a high-throughput intelligent memory fabric to avoid any memory bottleneck when transferring data. The Akida IP Platform by BraiChip also utilises DMA in a similar manner. It also has a runtime manager that manages all operations of the neural processor with complete transparency and can also be accessed through a simple API.

Other features of these platforms include the multi-pass processing, which is processing multiple layers at a time, making the processing very efficient. One ability of the device is the smaller footprint integration to make fewer nodes in a configuration and go from a parallel type execution to a sequential type execution. This means latency.

But a huge amount of that latency comes from the fact that the CPU gets involved every time a layer comes in. As the device processes multiple layers at a time, and the DMA manages all activity itself, latencies are significantly reduced.

Consider an AV, which requires the data to be processed from various sensors at a time. The TDA4VM from Texas Instruments Inc. comes with a dedicated deep-learning accelerator on-chip that allows the vehicle to perceive its environment by collecting data from around four to six cameras, a radar, lidar, and even from an ultrasonic sensor. Similarly, the Akida IP Platform can do larger networks on a smaller configuration simultaneously.

Glimpse into the market

These devices have a huge variety of scope in the market. As these are referred to as event based devices, they have an application sector. For example, Google's Tensor processing units (TPUs), which are application-specific integrated circuits (ASICs), are designed to accelerate deep learning workloads in their cloud platform. TPUs deliver up to 180 teraflops of performance, and have been used to train large-scale machine learning models like Google's AlphaGo AI.

Similarly, Intel's Nervana neural network processor is an AI accelerator designed to deliver high performance for deep learning workloads. The processor features a custom ASIC architecture optimised for neural networks, and has been adopted by companies like Facebook, Alibaba, and Tencent.

Qualcomm's Snapdragon neural processing engine AI accelerator is designed for use in mobile devices and other edge computing applications. It features a custom hardware design optimised for machine learning workloads, and can deliver up to 3-terflop performance, making it suitable for on-device AI inference tasks.

Several other companies have already invested heavily in designing and producing neural processors that are being adapted into the market as well, and that too in a wide range of industries. As AI, neural networks, and machine learning are becoming more mainstream, it is expected that the market for neural processors will continue to grow.

In conclusion, the future of neural processors in the market is promising, although many factors may affect their growth and evolution, including new technological developments, government regulations, and even customer preferences. EFY

This article, based on an interview with Nandan Nayampally, Chief Marketing Officer, BrainChip, has been transcribed and curated by Jay Soni, an electronics enthusiast at EFY